
Robust Low-Light Image Enhancement via Generative Priors: A CSCE753 Project Midterm Report

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1 Introduction

This midterm report builds on our initial proposal for robust low-light image enhancement using generative priors. In this project, we compare two unsupervised low-light enhancement approaches—one based on a multi-modal data generation pipeline (incorporating depth estimation, dynamic degradation, and image captioning) and the other employing a latent-Retinex diffusion model. The updated scope emphasizes tackling domain adaptation challenges and integrating multi-modal data, particularly by extending evaluations to a plant dataset for downstream phenotyping tasks.

To address reproducibility challenges observed in recent studies, we have developed a robust, containerized codebase using Docker Compose. This setup encapsulates dependencies and standardizes environments across platforms, ensuring that experiments run consistently. Additionally, we integrate FastAPI for rapid prototyping of model interfaces and leverage workflow orchestration tools (such as Celery, Airflow, and MLflow) to streamline experiment management and performance monitoring.

Our approach not only facilitates a comprehensive comparison between the selected methods from [1] and [2], but also lays the groundwork for future research in plant phenotyping analysis. By combining state-of-the-art low-light enhancement techniques with advanced reproducibility and deployment strategies, this work aims to bridge the gap between academic research and practical, real-world applications.

2 Literature Review

Recent advances in low-light image enhancement have leveraged diffusion models to address the challenges posed by unpaired training data and diverse illumination conditions. In **LightenDiffusion** [2], the authors introduce an unsupervised framework based on latent-Retinex diffusion models. Their method decomposes the enhancement process into latent components, allowing for effective restoration of underexposed images without relying on paired datasets. This approach not only improves the visual quality of dark images but also demonstrates promising gains in downstream tasks, providing a strong case for its application in real-world scenarios where obtaining paired training data is challenging.

Conversely, **Light the Night** [1] proposes a multi-condition diffusion framework specifically designed for the autonomous driving domain. This work addresses the broader problem of dark image enhancement by incorporating both supervised and unsupervised techniques. It tackles the inherent difficulties of processing images with severe underexposure and overexposure, thereby enhancing visual perceptibility under a wide range of lighting conditions. The framework's ability to adapt to the unique challenges of real-world nighttime driving scenes informs our project's focus on domain adaptation and multi-modal integration, particularly in scenarios that extend beyond conventional datasets.

Together, these studies highlight the importance of unsupervised learning strategies in overcoming data limitations, and they underscore the potential of diffusion-based methods in achieving robust low-light enhancement. Their insights directly inform our approach, as we aim to compare these models on diverse datasets and extend their applicability to new domains, such as plant imagery, where multi-modal cues and domain adaptation are crucial.

3 Proposed Method

Our primary objective is to reproduce and compare the two recent unsupervised low-light image enhancement frameworks: **Light the Night** [1] and **LightenDiffusion** [2]. We will first implement each model’s training and inference pipeline on standard low-light datasets (e.g., LOL, LSRW), collecting baseline results using metrics such as PSNR, SSIM, LPIPS, and NIQE. These baselines will serve as a benchmark for subsequent experiments aimed at evaluating inference speed, memory consumption, and robustness to varying lighting conditions.

The **Light the Night** method [1] includes a multi-condition diffusion mechanism and a data generation pipeline that integrates LiDAR or depth information, a dynamic degradation process, and textual guidance (via captions) for reward modeling. This multi-modal strategy may be valuable for improving domain adaptation to real-world environments, including plant imagery. Meanwhile, the **LightenDiffusion** approach [2] employs a latent-Retinex diffusion framework that decomposes low-light images into content and illumination layers, potentially offering a more principled way to remove noise and enhance details without paired training data.

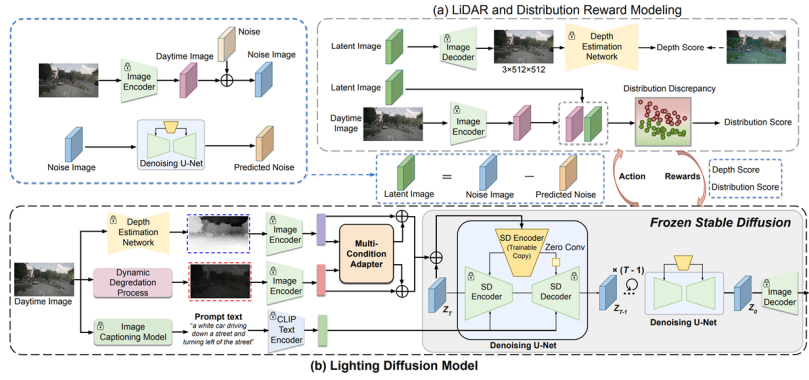


Figure 1: Framework for **Light the Night** [1].

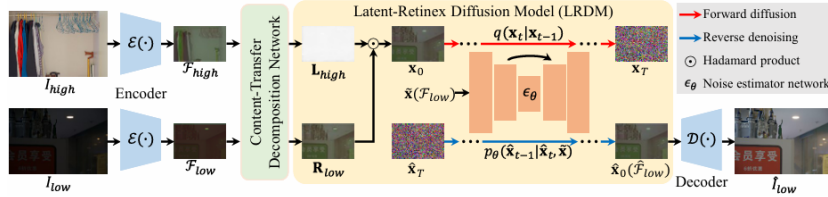


Figure 2: Framework for **LightenDiffusion** [2].

Beyond direct comparisons, we plan to explore potential *hybrid* strategies, such as incorporating the multi-modal reward design from [1] into the latent-Retinex structure of [2], if time and resources permit. Additionally, we will investigate how both approaches perform on a plant dataset, focusing on downstream segmentation tasks for phenotyping. This step will help assess the models’ adaptability to specialized domains and guide potential optimizations (e.g., fine-tuning network components, modifying the reward function, or introducing domain-specific priors).

Overall, our methodology remains flexible: we aim to validate each framework independently, measure their efficiency and performance on multiple datasets, and explore combining or refining their core ideas. Such a modular approach ensures that, even if resource constraints limit full-scale

integration, the comparative study will yield valuable insights for low-light enhancement in both general and domain-specific contexts.

4 Preliminary Investigations

Our initial investigations have revealed several challenges and opportunities that guide our ongoing work. A primary observation is the lack of standardized software engineering best practices in current research implementations. Many state-of-the-art low-light enhancement projects, including the ones in [1] and [2], exhibit issues such as unclear code structures—where methods perform multiple tasks unnecessarily—and environment setup inconsistencies. For example, the **LightenDiffusion** codebase lists requirements that work only with Python 3.8, without a clear explanation of these dependencies, leading to reproducibility challenges.

To address these issues, we have begun developing a repeatable, containerized codebase that bridges the gap between model development and production environments. Our approach leverages Docker and Docker Compose to encapsulate dependencies and ensure that experiments run identically across different platforms. Additionally, we are integrating FastAPI for rapid prototyping of model interfaces, and employing workflow orchestration tools such as Celery, Airflow, and MLflow to manage experiment pipelines and monitor performance. This setup is expected to not only streamline the development process but also greatly improve the reproducibility of results—a crucial factor when comparing the two recent diffusion-based approaches.

Preliminary experiments have focused on establishing these containerized environments and running simplified pipelines for both [1] and [2]. Early tests confirm that a modular design, which separates data ingestion, model training, and evaluation, can significantly reduce setup overhead and ease the troubleshooting process. These findings provide a strong justification for our methodological emphasis on reproducibility and rapid prototyping. They also underscore the potential of our approach to be extended beyond this project.

Figure 3 [3] illustrates our proposed microservice-oriented architecture, in which requests from a client pass through an NGINX reverse proxy and into a FastAPI application. The FastAPI server enqueues tasks into a task broker (e.g., Redis or RabbitMQ), which the Celery worker consumes asynchronously. A dedicated results backend ensures that outputs remain accessible even if individual components need to restart. By separating the core *business logic* (e.g., low-light image enhancement) from the rest of the pipeline, we can easily integrate additional modalities or adapt to new datasets without disrupting the entire system. This modular design fosters both scalability and reproducibility, as each service runs in its own container and can be independently updated or replaced.

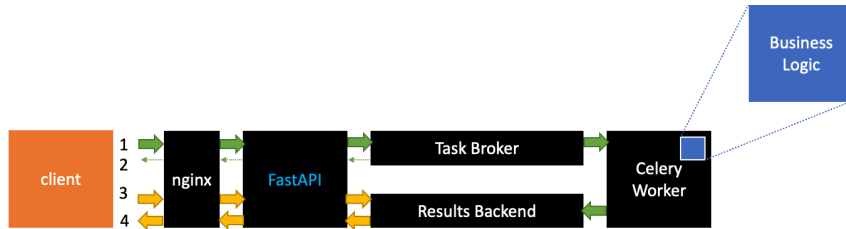


Figure 3: Proposed containerized architecture using Docker Compose and workflow orchestration tools for reproducible deployment. [3]

Looking ahead, this project is intended as a stepping stone towards a broader research agenda in plant phenotyping analysis. The methodologies developed here will facilitate the integration of multi-sensor data—including Red, Green, RedEdge, and NIR channels—to improve data generation processes for low-light conditions. Although integrating such diverse modalities presents its own set of challenges, the insights gained from our preliminary work with the selected models will inform future experiments aimed at domain adaptation and multi-modal fusion.

It is important to note that the two models chosen for this study come from very recent publications ([1] from April 2024 and [2] from July 2024). Both approaches claim superior performance over previous methods, yet they are tailored for slightly different applications. The flexibility of our proposed pipeline allows us to either optimize one of these approaches individually or explore a

hybrid strategy—combining the multi-modal and reinforcement learning elements from [1] with the latent-Retinex diffusion structure of [2]. A list of targeted experiments has been drafted, contingent on available time and resources, to systematically evaluate performance improvements, domain adaptability, and computational efficiency.

Overall, these preliminary investigations have not only validated our technical strategy but also shaped the future direction of our research. By emphasizing reproducibility and a modular, containerized approach, we aim to create a robust platform that supports extensive comparative studies and facilitates future innovations in low-light image enhancement and plant phenotyping analysis.

References

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